

MICHAEL WIECK-SOSA

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EDUCATION

Carnegie Mellon University | PhD in Statistics | Advisor: [Aaditya Ramdas](#) May 2027

- GPA: 3.98/4.00 | Awards: selected by faculty for the DeGroot-Goel Fellowship (1 Statistics PhD student per year) in 2026
- Thesis topics: prediction with transformers, estimation of time series models, and inference for dependence relationships

University of Illinois at Urbana-Champaign | MS in Statistics May 2022

- GPA: 3.95/4.00 | Awards: two years of guaranteed assistantships with full tuition waiver and stipend

Fordham University | BS in Mathematics with Minors in Computer Science and Economics May 2020

- GPA: 3.77/4.00 | Awards: *magna cum laude* | GRE: 170/170 Quantitative, 163/170 Verbal, 4.5/6.0 Writing

PROGRAMMING LANGUAGES AND TOOLS

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- Python expert: extensive experience with NumPy, pandas, Polars, scikit-learn, statsmodels, and PyTorch
 - R expert: extensive experience with dplyr, Rcpp, xts, zoo, caret, mgcv, glmnet, parallel, and ggplot2
 - SQL, Git, Bash expert: extensive experience with data management, version control, and Linux
 - C++ proficient: used for courses in Data Structures and Algorithms
 - q/kdb+ proficient: used during internship to work with high-frequency financial data

COURSEWORK

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- **Statistics:** Advanced Statistical Theory, Intermediate Statistics, Mathematical Statistics, Advanced Time Series Analysis, Regression Analysis, Computational Statistics
 - **Computer Science:** Algorithms, Data Structures, Theory of Computation, Operating Systems, Computer Architecture, Artificial Intelligence, Machine Learning, Data Mining for Listening to the Social Universe
 - **Math:** Stochastic Calculus, Measure-theoretic Probability, Functional Analysis, Measure Theory, Interacting Particle Systems, Geometric Flows, Differential Geometry, Lie Groupoids & Lie Algebroids, Topology, Abstract Algebra, Numerical Analysis, Numerical Linear Algebra, Real Analysis, Differential Equations, Linear Algebra, Mathematical Modeling

DOCTORAL RESEARCH

Transformer-based Foundation Models and Deep Neural Networks for Predicting Nonstationary Sequences | Aaditya Ramdas

- Developing methods for predicting nonstationary sequences using state-of-the-art neural network architectures

Invariant Prediction for Nonstationary Sequences | Aaditya Ramdas

- Creating a statistical method that can be used to make machine learning models more robust to nonstationarity

Sequence Analysis Methods for Large Language Models | Leif Weatherby, Tyler Shoemaker, and Cosma Shalizi

- Developing method for analyzing sequences output by large language models based on advanced stochastic process theory

Dynamic Variable Importance for Foundation Models | CMU Undergraduate Student and Chad Schafer

- Creating a method to measure the importance of time series input into foundation models like Chronos-2 and TabPFN-TS

Goodness-of-fit Testing and Confidence Sets through Random Features | Cosma Shalizi

- Developing a framework for uncertainty quantification and testing whether the law of a stochastic process is in a model class

Simulation-based Estimation of Dynamic Models by Matching Random Features | Cosma Shalizi

- Creating a likelihood-free method for estimating the parameters of complicated models for time series

State-space Methods for Epidemic Data | Michel F. C. Haddad and Aaditya Ramdas

- Developing a state-space model that links population viral load, mobility, vaccination coverage, and variant frequency

Conditional Independence Testing for Nonstationary Time Series | Michel F. C. Haddad and Aaditya Ramdas

- Creating a conditional independence test based on regression that is robust to nonstationarity and temporal dependence

GRADUATE RESEARCH ASSISTANTSHIPS

Carnegie Mellon University | [NSF Grant](#) | PI: Cosma Shalizi

May 2024-Present

- Developing theory, methods, and software in Python and R for estimating and inferring the parameters of dynamic models
- Using optimization algorithms to optimize highly non-convex objective functions to fit complicated models for time series
- Applied methods to time series models, state-space models, SDEs, ODEs, and dynamical systems with observational noise

University of Illinois at Urbana-Champaign | FORWARD Data Lab | Computer Science Department

Jan. 2021-May 2021

- Discovered patterns in cross-platform dynamics on Twitter, Facebook, and Reddit with Hawkes processes using Python

National Center for Supercomputing Applications | Innovative Software and Data Analysis Group

Sept. 2020-May 2022

- Built confidence bands for trends in concentrations and fluxes of chemicals to measure changes in water quality over time
- Used parallel computing to construct these confidence bands for 1000+ locations in the U.S. and made visualizations using R

INDUSTRY INTERNSHIPS

J.P. Morgan | Quantitative Research | Markets Summer Associate | Equity Derivatives and Macro

June 2023-Aug. 2023

- Worked with macro traders and quants on a hedging method for derivatives portfolios via multi-period optimization
- Collaborated with energy derivatives traders on improving the statistical techniques used in a new trading strategy

J.P. Morgan | Quantitative Research | Markets Summer Associate | Equity Derivatives

June 2022-Aug. 2022

- Developed method for adaptively selecting parameters of trade execution algorithms based on real-time data
- Built pipeline for processing market microstructure data using q/kdb+ and making predictions using scikit-learn in Python
- Used Bayesian optimization and walk-forward cross-validation to tune the hyperparameters of machine learning models

RESEARCH INTERNSHIPS

MIT Lincoln Lab | Group 38

May 2021-July 2021

- Implemented optical tracking methods for tracking objects in outer space and ran simulations to compare approaches

Corteva Agriscience | Research & Development Division

June 2020-Aug. 2020

- Analyzed environmental changes based on time series of 50+ weather and soil features from 1,000+ sensor sites in the U.S.
- Used spatiotemporal Gaussian mixture models, dimensionality reduction, and clustering techniques with Python
- Built an interactive Dash application to visualize my statistical analysis of how environments changed over space and time

Fordham University | Robotics and Computer Vision Lab | Computer Science Department

Aug. 2019-March 2020

- Created a real-time tracking system for the Kihansi spray toad in collaboration with zoologists using deep learning in Python

TEACHING ASSISTANTSHIPS

- For the MS in Computational Finance program at CMU: Simulation Methods for Option Pricing, Financial Time Series, Financial Data Science I and II, and the Machine Learning Capstone Project
- For the MS in Data Science program at CMU: Time Series and Experimental Design
- For the BS in Statistics and Statistics/Machine Learning programs at CMU: Advanced Data Analysis

PROFESSIONAL SERVICE

- 2026: reviewer of one paper for the Journal of the Royal Statistical Society, Series B (Statistical Methodology)
- 2024-2025: chair of the Statistics PhD student committee at Carnegie Mellon University focused on organizing career events

TALKS AND POSTERS

- 2026: in-person talk on conditional independence testing at the International Workshop in Sequential Methodologies at American University
- 2025: in-person talk on simulation-based inference by matching random features at the Statistical Methods for the Physical Sciences (STAMPS) Research Center's local meeting at Carnegie Mellon University

- 2025: poster on simulation-based inference by matching random features at the Workshop on Neural Simulation-based Inference at Carnegie Mellon University
- 2025: online talk on variable selection for large-scale forecasting for nonstationary time series to researchers at Amazon
- 2025: online talk on conditional independence testing at the Virtual Time Series Seminar's Workshop for Junior Researchers
- 2024: poster on conditional independence testing at the NBER-NSF Time Series Conference at University of Pennsylvania
- 2024: in-person talk on regression for time series at Statistics and Machine Learning Group at Carnegie Mellon University